

Halide-sensitive fluorescence assays

- Measure quenching of yellow fluorescent protein (YFP) as halide concentration increases or decreases within the cell
- Can use NaI as a channel substrate for CFTR → if CFTR is intact, intracellular iodide concentration increases and less fluorescence would be detected
 - If CFTR is not intact, intracellular iodide concentration decreases and we would see more fluorescence
- To track fluorescence:
 - Hardware:
 - Fluorescence microplate readers (the fancier ones cost between \$20,000 to \$100,000)
 - Flow cytometer
 - Assay measurements:
 - Measure the initial fluorescence before adding the NaI
 - Decide on how many timepoints we'll collect a measurement
 - At each timepoint, measure the fluorescence and create a percentage relative to the initial timepoint/fluorescence
- Cell lines: use epithelial cell lines to mimic CFTR's behaviour in the lungs
 - Example: Calu-3 cells, FRT (Fischer Rat Thyroid) cells
 - If the chosen cell line does not already have CFTR ion channel or YFP, these will have to be transfected into the cells
- Control options
 - DMSO (can help us establish an iodide baseline influx rate)
 - Forskolin (a CFTR channel activator)
 - CFTR(inh)-172 (a CFTR channel blocker)
- Article examples for more information:
 - Xia, S., Di Paola, M., Jones, N. L., & Bear, C. E. (2022). A Fluorescence-based Assay of Membrane Potential for High-throughput Functional Study of Two Endogenous Ion Channels in Two Epithelial Cell Lines. *Journal of visualized experiments : JoVE*, (184), 10.3791/63528. <https://doi.org/10.3791/63528> (<https://pubmed.ncbi.nlm.nih.gov/35815993/>)
 - Smith, E., Giuliano, K. A., Shumate, J., Baillargeon, P., McEwan, B., Cullen, M. D., Miller, J. P., Drew, L., Scampavia, L., & Spicer, T. P. (2017). A Homogeneous Cell-Based Halide-Sensitive Yellow Fluorescence Protein Assay to Identify Modulators of the Cystic Fibrosis Transmembrane Conductance Regulator Ion Channel. *Assay and drug development technologies*, 15(8), 395–406. <https://doi.org/10.1089/adt.2017.810> (<https://pubmed.ncbi.nlm.nih.gov/29172645/>)